WORKED SOLUTION

STEP I 2002, Question 14

Probability and Statistics • Geometric Series in Probability

STEP I 2002, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/02-s1-q14

The key idea

Condition on how many numbers I punch in altogether. Once that number is fixed, the time between the alarm going off and my getting the money is fixed too, so the police probability is just $1 - \mathrm{e}^{-\lambda t}$ for that time. Averaging over the number of attempts gives the answer. In both plans the averaging turns into a geometric series in $q = \mathrm{e}^{-\lambda/r}$, so it's worth naming q early.

Solution

Punching in order

Let K be the position of my number in the list $1, 2, \ldots, n$. Then K is the number of attempts I make, and $P(K = k) = \frac{1}{n}$ for $k = 1, \ldots, n$.

Suppose $K = k$. I punch numbers at a rate of r per minute, so successive numbers are $1/r$ minutes apart. Measure time from the first number I punch.

- If $k = 1$, the first number is right, no wrong number is ever entered and the police are never called. The probability that they arrive first is $0 = 1 - \mathrm{e}^{0}$.
- If $k \geq 2$, the first number is wrong, so the alarm is raised at time 0, and the correct number is the kth, entered at time $(k-1)/r$. The police arrive before I get my money if they arrive within $t = (k-1)/r$ minutes, which has probability $1 - \mathrm{e}^{-\lambda(k-1)/r}$.

The formula $1 - \mathrm{e}^{-\lambda(k-1)/r}$ therefore covers every k, including $k = 1$. Let A be the event that the police arrive before I get my money. By the law of total probability,

$$P(A) = \sum_{k=1}^{n} P(K = k)\, P(A \mid K = k) = \sum_{k=1}^{n} \frac{1 - \mathrm{e}^{-\lambda(k-1)/r}}{n},$$

as required.

To simplify, put $q = \mathrm{e}^{-\lambda/r}$. Since $\lambda, r > 0$ we have $0 < q < 1$, and $\mathrm{e}^{-\lambda(k-1)/r} = q^{k-1}$. So

$$P(A) = 1 - \frac{1}{n}\sum_{k=1}^{n} q^{k-1} = 1 - \frac{1-q^n}{n(1-q)}$$
$$= 1 - \frac{1-\mathrm{e}^{-\lambda n/r}}{n\left(1-\mathrm{e}^{-\lambda/r}\right)},$$

summing the geometric series, which is valid because $q \neq 1$.

Punching at random

Now each attempt is a fresh number chosen uniformly from $1, \ldots, n$, independently of earlier attempts (I don't remember what I've tried). So each attempt is correct with probability $\frac{1}{n}$, independently. If N is the number of attempts up to and including the correct one, then N is geometric:

$$P(N = k) = \frac{1}{n}\left(\frac{n-1}{n}\right)^{k-1}, \qquad k = 1, 2, 3, \ldots$$

Ask yourself Does the timing argument still work when numbers can be repeated?

Yes. The police are alerted at the first wrong number, and if $N = k \geq 2$ that is the very first attempt, at time 0. The money comes at the kth attempt, at time $(k-1)/r$. Repeats change only the distribution of N, not the timing. So exactly as before, $P(A \mid N = k) = 1 - q^{k-1}$, and this is also right for $k = 1$.

Write $p = \frac{n-1}{n}$. Then

$$P(A) = \sum_{k=1}^{\infty} \frac{1}{n} p^{k-1}\left(1 - q^{k-1}\right) = \frac{1}{n}\sum_{k=1}^{\infty} p^{k-1} - \frac{1}{n}\sum_{k=1}^{\infty}(pq)^{k-1}$$
$$= \frac{1}{n}\cdot\frac{1}{1-p} - \frac{1}{n}\cdot\frac{1}{1-pq}.$$

Splitting the sum is fine because both series converge: $0 \leq p < 1$ and $0 < pq < 1$. The first term is $\frac{1}{n}\cdot n = 1$, which is just the statement that N is certainly finite. For the second,

$$n(1-pq) = n - (n-1)q,$$

so

$$P(A) = 1 - \frac{1}{n - (n-1)\mathrm{e}^{-\lambda/r}},$$

as required.

Checks

- With $n = 1$ there is never a wrong number, and both formulae give 0: the first because $1 - q^1 = 1 - q$, the second because the denominator is 1.

- As $\lambda \to \infty$ (the police arrive at once), $q \to 0$ and both answers tend to $1 - \frac{1}{n}$. That's the probability that the first attempt is wrong, as it should be.
- Random punching should be worse. For $n = 5$, $\lambda = 0.7$, $r = 3$ we get 0.3382 in order and 0.4543 at random. A simulation with 2×10^5 trials of each gave 0.339 and 0.456.

Common slips

- Using time k/r instead of $(k-1)/r$. The clock starts at the first (wrong) number, so the kth number comes $k - 1$ gaps later.
- In the random case, giving each k a probability of $\frac{1}{n}$, or stopping the sum at n. With repeats allowed, N is geometric and unbounded.

Beyond STEP

The police arrival time here is an exponential random variable, and the calculation is $P(T < X)$ for a random X, found by conditioning on X. When X is geometric the answer has a neat closed form because $E\left(q^X\right)$ is the probability generating function of X evaluated at q. Indeed both answers are $1 - E\left(q^{N-1}\right)$. Generating functions and the exponential distribution are central to first-year probability. Grimmett and Welsh, *Probability: An Introduction*, covers both well.

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